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Determining Reservoir Fluid Properties from Unconventional Reservoir Core Analysis

Anuj Gupta, Stacey Althaus, and Shannon Eichmann, Aramco Americas, Houston, Tx, USA; Rabah Mesdour, Saudi Aramco, Dhahran, Saudi Arabia

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Abstract

Determination of reservoir fluid type for unconventional reservoirs is quite challenging but very important for reservoir development planning. This work presents a method for predicting the reservoir fluid formation volume factor, B_o and fluid type from routinely conducted core analyses. The proposed method utilizes the gas, oil, and water saturations measured from crushed core samples. The saturations represent the percentage of pore volume occupied by each fluid and are used to calculate the "Apparent formation volume factor" of reservoir fluid which, in turn, allows fluid classification as black oil, volatile oil, gas condensate, or gas. The analysis also allows calculation of important characteristics such as gas-oil ratio/condensate-gas ratio for the candidate reservoir fluid. To demonstrate the method, it was applied to various unconventional reservoir samples across several wells and reservoir intervals. Oil formation volume factor, solution gas oil ratio, and condensate gas ratio were calculated from experimental results and used to determine the reservoir fluid type. The predicted fluid types are compared to those observed from other industry standard methods. The workflow, equations, and evaluation criteria used are shared and results obtained from using fluid saturations from both the gas research institute (GRI) method and the mudrock reservoir properties (MRP) method were utilized to show the versatility of the methodology. The results demonstrate that the method offers an efficient approach for estimating reservoir fluid type, and gas-oil ratio/condensate-gas ratio using routinely available core analysis.

Introduction

Unconventional Reservoirs around the world are characterized by very low primary recovery. Published estimates indicate that as much as 90% of original oil may be left in the reservoir at the end of primary recovery (Jia et. al, 2019). This suggests that there is a large potential for enhancing the recovery using suitable technology. However, in order to utilize this potential, we must understand what resources are left downhole.

Most unconventional shale reservoirs are known to be highly laminated with minimal communication between different stacked geological formations even when they are adjacent to each other vertically. Because of such vertical isolation, formations may contain different fluid types that could range from black

oil to dry gas, depending on the maturity of the source rocks. Some of these formations have higher oil/gas content and can be more productive than other formations in the reservoir. As a result, these formations deplete differently and the amount of residual oil remaining after primary production may be different in each formation. Determining the fluid type associated with each geological formation and the expected gas-oil ratio (GOR) and condensate-gas ratio (CGR) is of great importance for determining the expected fluid production and evaluation of the lifetime economic value of target reservoir, effective field development planning, and design of surface facilities.

For conventional reservoirs, the fluid type, GOR, and CGR are traditionally determined by collecting a sub-surface sample of the reservoir fluid. Alternatively, a representative sample may be collected at the surface by recombining produced hydrocarbons in an observed producing ratio. Such representative reservoir fluid samples or sub-surface samples may be used to measure the hydrocarbon properties, including oil formation volume factor (B_o), GOR, CGR, bubble point pressure, dew point pressure and the like, as well as to classify the reservoir fluids (e.g., black oil, volatile oil, retrograde condensate, wet gas, dry gas).

Such methods are cost prohibitive and can be inaccurate for unconventional reservoirs because the fluid type between various contributing geological formations can change dramatically. In order to obtain useful and/or accurate analysis of an unconventional reservoir, multiple strategically selected samples may be required when evaluating fluid production via conventional methods and systems.

This paper demonstrates a method to estimate B_{oi} , fluid type, GOR, and CGR in unconventional reservoirs from fluid saturations obtained from crushed rock samples during routine core analyses. Predictions of fluid types from the method are compared to observed fluid types for demonstration purposes.

Methods

Samples

Unconventional source rock samples representing a variety of rock types across many wells were obtained for evaluation. Samples of different maturities were obtained to ensure that the sample set covered a wide range of expected values. Sample selection was done using x-ray tomography (CT) to determine the most representative region of the samples.

Petrophysical properties tests

The proposed method utilizes the laboratory test results that are conducted as part of evaluating the properties of shale oil/gas reservoirs. The most common of these tests is referred to as the Gas Research Institute (GRI) Method and another similar test procedure is referred to as the Mudrock Reservoir Properties (MRP) Method. The measurements of relevance, from the GRI, MRP and any other similar tests, are the saturations of gas, oil, and water present in the tested samples.

GRI method. GRI was accomplished using the method described in (Luffel and Guidry 1992, Guidry et al. 1996). In brief, a sample is cut from the core, and a representative piece is taken from the center of the core to determine bulk density via mercury immersion. The remaining sample is crushed, and a portion of the crushed material is then run through the Dean-Stark distillation process (Dean and Stark 1920) with toluene to remove the fluids from the sample. The sample is then dried, and a grain volume measurement is conducted. The results obtained from GRI include residual water saturation, S_{w_lab} and residual oil saturation, S_{o_lab} .

MRP method. In brief, the MRP method (Suarez-Rivera et al. 2016) combines thermal and solvent extraction (i.e., Dean-Stark (Dean and Stark 1920), Karl Fischer (Fischer 1935), and Accelerated Solvent Extraction (ASE) (Richter et al. 1996)) methods with other core analyses to measure fluid saturations (i.e., oil and water saturations) and other properties from crushed rock samples. From such analyses, values

similar to those provided by GRI analyses are obtained, the important ones for this publication being water saturation, S_{w_lab} and oil saturation, S_{o_lab} .

Estimation of Reservoir Oil Formation Volume Factor and Fluid type. The results from the GRI or MRP analysis are used to determine an apparent reservoir oil formation volume factor, B_{oi} . B_{oi} is the ratio of the volume of oil at downhole reservoir temperature and pressure to the oil volume at standard temperature and pressure. This ratio will give insight into the fluid type. Although the temperature and pressure play a role in the volume of the fluids, the quantity of dissolved gas in the reservoir is the most important factor in determining the B_{oi} . This is crucial in unconventional reservoirs where the quantity of solution gas varies significantly compared to conventional reservoirs.

If the original reservoir fluid in the core samples was black oil or volatile oil, the original oil saturation, S_{oi} , in the reservoir can be calculated using Equation 1.

$$S_{oi} = (1 - S_{wi}) \quad (1)$$

where S_{wi} is the initial water saturation. This is based on an assumption that the reservoir pressure is above the bubble point pressure and the reservoir is undersaturated. There is no free gas saturation in the pore space at the initial reservoir condition for an undersaturated reservoir.

The retrieval of the core samples from the reservoir causes the pressure to drop below the bubble point pressure and the subsequent expulsion of the solution gas. Since the reservoir permeability is ultra-low in unconventional reservoirs, it is reasonable to assume that only solution gas is able to escape the pore space and when the core is brought to surface pressure and temperature conditions, the residual oil saturation will be reduced to a value comparable to that of stock tank oil. Therefore, the ratio of original oil saturation to the oil saturation measured in the retrieved core sample may be used as an approximation to the oil formation volume factor, B_{oi} , as calculated by Equation 2.

$$B_{oi} = \frac{1 - S_{w_lab}}{S_{o_lab}} \quad (2)$$

Where S_{w_lab} is the water saturation and S_{o_lab} is the oil saturation both measured by GRI or MRP on crushed rock samples. It should be noted that this assumes very little change in the water saturation between downhole and lab sampling.

For a dry gas, wet gas, or gas condensate reservoir, the original reservoir fluid is presumed to be above the dew point pressure and for most practical purposes can be considered to be a gas saturation. Therefore, there is no oil saturation in such reservoirs at the original reservoir conditions. As the core is retrieved from the reservoir to the surface, the pore pressure drops below the dew point pressure and condensate (liquid) drop off is expected for wet gas or retrograde condensate reservoirs. In these cases, owing to the ultra-low unconventional reservoir permeability, it is reasonable to assume that only gas is able to escape the pore space leaving any condensate trapped in the pore space as residual oil. Here, when the core is brought to surface pressure and temperature conditions, the residual oil saturation may be comparable to the maximum liquid drop off typically measured in reservoir fluid analyses. The residual oil saturation for such wells may be compared to the maximum liquid drop off expected in constant volume depletion tests.

According to fluid classification in (Lake and Holstein), the following are an indicator of reservoir fluid type based on the available measurements of gas, oil and water saturation to determine formation volume factors:

- Black oil: If apparent oil formation volume factor is less than 1.5
- Volatile oil: If apparent oil formation volume factor is greater than 1.5 and less than 3.0.
- Gas condensate: If apparent oil formation volume factor is greater than 3.0.

Results and Discussion

The method to estimate B_{oi} from fluid saturations measured from routine core analyses is applied to a set of ten samples from various locations. Columns 2 and 3 in Table 1 shows example oil and water saturations measured by the GRI method. These data are used with Equation 2 to calculate the Apparent Oil Formation Volume Factor (B_{oi}) listed in Column 4 of Table 1. Column 5 lists the fluid classification for each sample using the criteria listed in the preceding (Lake and Holstein). Columns 4 and 5 show that the method predicts a range of B_{oi} values and fluid types as expected from the range of thermal maturities (data not shown) in the sample set. The next section discusses how the gas oil ratio (GOR) and condensate gas ratio (CGR) (Columns 6 and 7) were calculated.

Table 1—Example data using oil and water saturations obtained from the GRI methods to estimate apparent oil formation volume factor, fluid type, GOR, and CGR.

Sample ID	Oil Saturation (S_{o_lab})	Water Saturation (S_{w_lab})	Apparent Oil Formation Volume Factor (B_{oi}) (RB/STB)	Fluid Type	GOR (SCF/ STB)	CGR (STB/ MMSCF)
1	61.94	29.33	1.14	Black Oil	54	-
2	67.88	8.09	1.35	Black Oil	394	-
3	52.51	11.94	1.68	Volatile Oil	870	1152
4	48	4.13	2.00	Volatile Oil	1295	772
5	40.57	3.74	2.37	Volatile Oil	1760	568
6	26.66	5.82	3.53	Gas Condensate	3105	322
7	15.87	6.67	5.88	Gas Condensate	5558	180
8	47.72	7.21	1.94	Volatile Oil	1217	822
9	9.26	12.41	9.46	Gas Condensate	8966	112
10	22.90	31.74	2.98	Volatile Oil	2490	402

Calculation of Reservoir Fluid Type Gas-Oil Ratio (GOR) and Condensate-Gas Ratio (CGR)

The amount of gas dissolved in a reservoir oil represented in standard cubic feet (SCF) per stock tank barrel (STB) is referred to as the solution gas-oil ratio (GOR). It is indicated by the symbol R_s in reservoir analysis and simulation. For a gas condensate reservoir, the stock tank barrels (STB) of condensate produced for every million standard cubic feet of gas (MMSCF) is referred to as condensate-gas ratio (CGR) and has units of STB/MMSCF. In terms of the units, it can be verified that

$$\text{GOR}(\text{SCF}/\text{STB}) = 1,000,000/\text{CGR}(\text{STB}/\text{MMSCF}) \quad (3)$$

The oil formation volume factor can be used to calculate the solution gas-oil ratio (GOR) using any of the numerous correlations available in the literature.

As an illustration, a correlation published in (McCain 1990) Eq. B-47, 48, is used to estimate the solution GOR based on B_{oi} values estimated by this method. First, a unitless parameter, CN, is calculated using Equation (4):

$$CN = \left(\frac{B_{oi} - 0.9759}{0.00012} \right)^{\frac{1}{1.2}} \quad (4)$$

Based on the results of CN obtained in Equation (4), solution GOR of the oil can be calculated using Equation (5):

$$GOR = \frac{CN - 1.25T}{\sqrt{\frac{SG_{gas}}{SG_{oil}}}} \quad (5)$$

where T is the reservoir temperature in degrees Fahrenheit, available from measurements during and after drilling (for the purposes of this paper we assumed reservoir temperature of 277°F), SG_{gas} is the specific gravity of produced gas with respect to air, available from produced gas from any off-set wells (assumed a value of 1.1 in the example calculations), SG_{oil} is the specific gravity of produced oil, available from measurement on produced oil from any offset wells (assumed a value of 0.78 in the example calculations).

This enabled calculation of GOR values listed in Column 6 of Table 1. Further, Equation 3 can be rearranged to calculate CGR values according to Equation (6):

$$CGR(STB/MMSCF) = 1,000,000 / GOR(STB/SCF) \quad (6)$$

CGR values calculated using equation (6) are listed in Column 7 of Table 1.

Additional examples are shown in Table 2 in which the B_{oi} , fluid type, GOR, and CGR are calculated using Eqn. 1 through Eqn. 6 for a wide range of samples analyzed using the MRP method. The oil formation volume factors are calculated using oil and water saturation from MRP using Eqn. 2 and the fluid type is determined from the criteria above (Lake and Holstein).

Table 2—Example using data from samples across a variety of unconventional reservoirs at different maturities. Columns 4-7 are calculated from the data in Columns 2 and 3 using Equations 2, 5, and 6. The saturations in this table were acquired via the MRP method.

Sample ID	Oil Saturation (S_o)	Water Saturation (S_w)	Apparent Oil Formation Volume Factor (B_{oi}) (RB/STB)	Fluid Type	GOR (SCF/ STB)	CGR (STB/ MMSCF)
1	17.90	8.02	5.14	Gas Condensate	4823	207
2	8.81	17.04	9.42	Gas Condensate	8931	112
3	38.00	6.71	2.45	Volatile Oil	1868	535
4	33.07	21.25	2.38	Volatile Oil	1778	563
5	42.84	8.26	2.14	Volatile Oil	1479	676
6	34.25	23.26	2.24	Volatile Oil	1604	623
7	24.65	14.21	3.48	Gas Condensate	3058	327
8	17.90	12.97	4.86	Gas Condensate	4539	220
9	36.05	7.56	2.56	Volatile Oil	2000	500
10	46.21	14.14	1.86	Volatile Oil	1112	899

From Table 1 and Table 2 it is clear that saturation data from multiple methods can be used to calculate the apparent B_{oi} and predict the fluid type for samples collected across a variety of locations and depths. However, these examples do not provide a comparison of how the method predictions compare what is observed from other data. For this comparison we calculated the average B_{oi} from this method and used it to predict the expected fluid type for several wells and compared it to PVT data (Table 3). This comparison shows that in 6 out of 7 cases the fluid measurement type from the method agrees with the PVT fluid type. Considering that some of the wells had variations in fluid type based on depth and formation contributing to the production, this gives us confidence that the method is accurately predicting the fluid type and B_{oi} .

Table 3—Comparison of fluid type across 7 wells. Fluid type from traditional PVT measurements compared to those predicted using the method described herein.

Well name	Fluid type PVT results	Fluid Type Method results
A	Gas Condensate	Gas Condensate
B	Volatile Oil	Volatile Oil
C	Gas Condensate	Gas Condensate
D	Gas Condensate	Gas Condensate
E	Gas Condensate	Gas Condensate
F	Gas Condensate	Volatile Oil
G	Gas Condensate	Gas Condensate

In summary, the method described in this paper can be summarized as shown in Figure 1. In short this can be written as: 1) Collect core samples from a reservoir, 2) Run GRI/MRP tests to determine oil and water saturations, 3) Use Equation 2 to calculate B_{oi} , 4) Use the B_{oi} to select the fluid type, 5) Use equation 5 to calculate GOR from B_{oi} , and 6) Use equation 6 to calculate CGR from GOR.

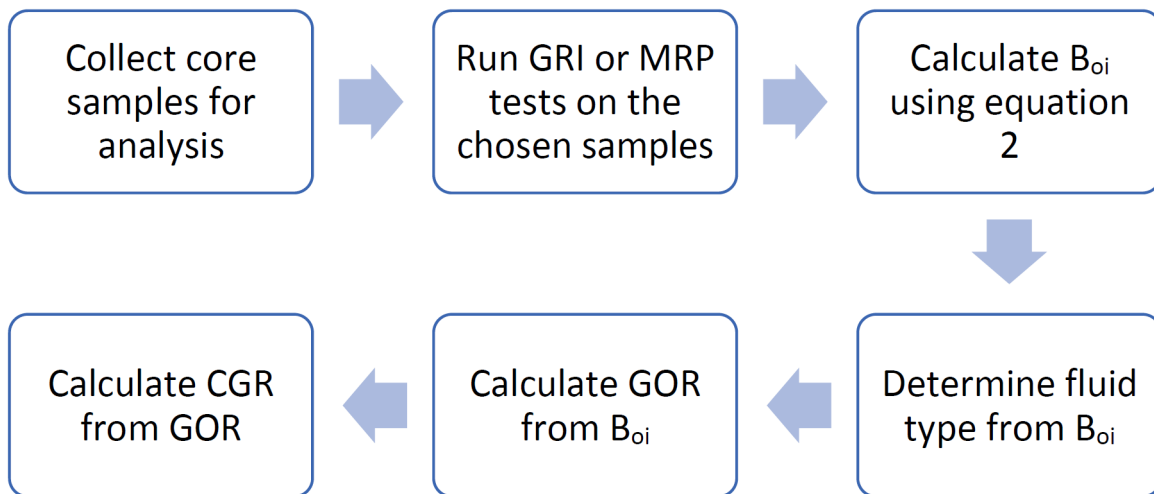


Figure 1—Workflow for the method described in this document to determine B_{oi} , fluid type, GOR, and CGR for unconventional reservoir samples.

Conclusion

In unconventional resources it is not straightforward to determine many reservoir parameters such as oil formation volume B_{oi} , fluid type, gas oil ratio (GOR), or condensate gas ratio (CGR). This paper demonstrated a method to estimate these properties using commonly measured petrophysical parameters, oil saturation and water saturation that are readily available from routine core analyses. The examples shown used fluid saturations obtained from commonly acquired GRI or MRP tests. Predicted fluid types were also compared to results from PVT analysis and were found to be in good agreement. This allows for an estimation of the expected fluid production, evaluation of the lifetime economic value of target reservoir, effective field development planning, and design of surface facilities. This can also be used to classify reservoirs and determine which may be a good possible target for enhanced recovery processes.

References

- Dean, E. W. and Stark, D. D. 1920. A Convenient Method for the Determination of Water in Petroleum and Other Organic Emulsions. *Journal of Industrial & Engineering Chemistry* 12 (5): 486-490. <https://doi.org/10.1021/ie50125a025>.

- Fischer, Karl. 1935. Neues Verfahren zur maßanalytischen Bestimmung des Wassergehaltes von Flüssigkeiten und festen Körpern. *Angew Chem* **48** (26): 394-396.
- Guidry, K., Luffel, D., and Curtis, J. 1996. Development of Laboratory and Petrophysical Techniques for Evaluating Shale Reservoirs. *Final Technical Report*, October 1986-September 1993.
- Lake, Larry W. and Holstein, Edward D. *Reservoir Engineering and Petrophysics*: Society of Petroleum Engineers.
- Luffel, D.L. and Guidry, F.K. 1992. New Core Analysis Methods for Measuring Reservoir Rock Properties of Devonian Shale. *Journal of Petroleum Technology* **44** (11): 1184-1190. <https://doi.org/10.2118/20571-PA>.
- McCain, W.D. 1990. *The Properties of Petroleum Fluids*: PennWell Books.
- Richter, Bruce E., Jones, Brian A., Ezzell, John L. et al 1996. Accelerated Solvent Extraction: A Technique for Sample Preparation. *Anal Chem* **68** (6): 1033-1039. <https://doi.org/10.1021/ac9508199>.
- Suarez-Rivera, Roberto, Von Gonten, W. D., Vaughn, K. et al 2016. Integration of Thermal and Solvent Extraction Methods for Improved Characterization of Hydrocarbon in Place and Producibility. Proc., SPE/AAPG/SEG Unconventional Resources Technology Conference. <https://doi.org/10.15530/URTEC-2016-2461786>.